

Last Time:

- RL summary

- POMDPs

lecture 5

421, FALL '26

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This Time:

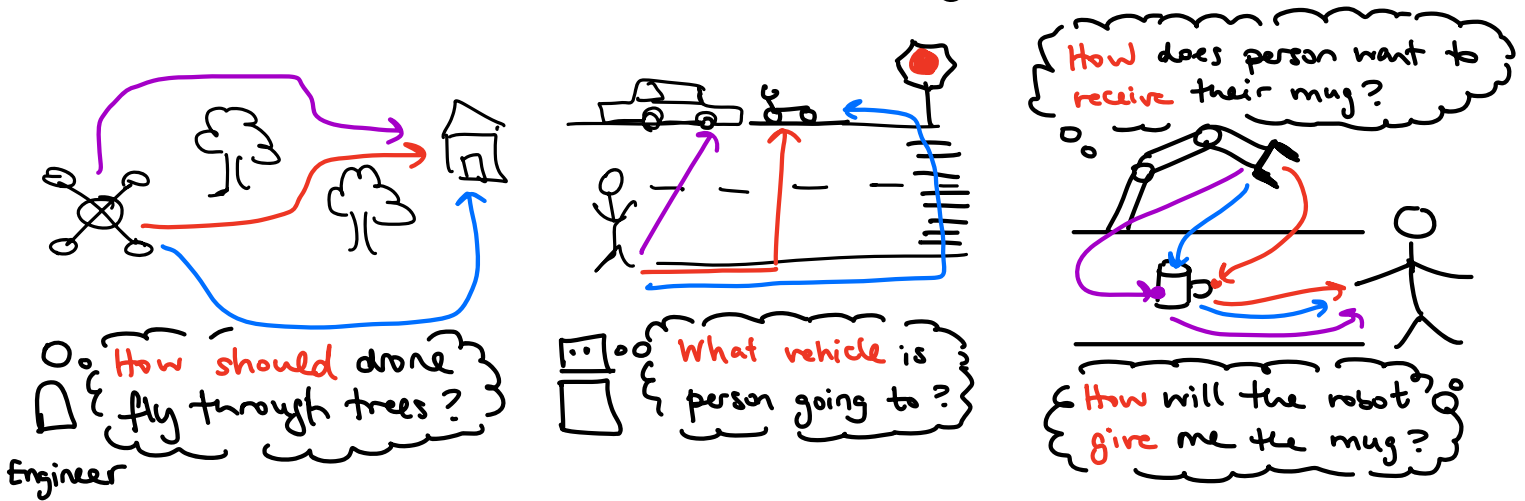
- Probability Recap

- Boltzmann Rationality

## Uncertainty:

We are starting to see that a lot of things in life are uncertain, including humans!

When we interact w/ other people, or the world, we may not know what their goals / intents / preferences / rewards are, even if we could perfectly observe their "physical" state!



Let's model this type of uncertainty! recap some basic probability theory along the way.

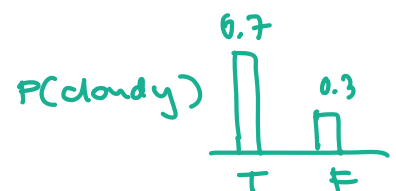
### Probability Recap

In probability theory, we want to make probabilistic assertions about possible "worlds" — possible human behavior, or environment responses ...

### NOTATION:

•  $P(X)$  probability of random variable  $X$

ex: cloudy  $\in \{\text{True}, \text{False}\}$   
    ↳ random variable



- $P(X, Y)$  probability of  $X$  and  $Y$ . Also called joint distribution  
↑  
"and"      ex.  $P(\text{cloudy}, \text{rain})$

- $P(X|Y)$  probability of  $X$  given  $Y$ . Also called conditional distribution.  
↑  
"given"      ex.  $P(\text{cloudy} | \text{rain})$

## USEFUL RULES

- Product Rule (or chain rule):

$$P(X, Y) = P(X|Y) \cdot P(Y) = P(Y|X) \cdot P(X)$$

ex. for it to be cloudy and rain (i.e.  $P(\text{cloudy}, \text{rain})$ ), we need it to rain (i.e.  $P(\text{rain})$ ) and we need it to be cloudy given the rain (i.e.  $P(\text{cloudy} | \text{rain})$ )

$$\begin{aligned} P(A, B, C, D) &= P(A | B, C, D) \cdot P(B, C, D) \\ &= P(A | B, C, D) \cdot P(B | C, D) \cdot P(C, D) \\ &= P(A | B, C, D) \cdot P(B | C, D) \cdot P(C | D) \cdot P(D) \end{aligned}$$

- marginalization (i.e. "summing out"):

$$P(X) = \sum_{y \in Y} P(X, y)$$

sums up probabilities of each possible value of a target variable (i.e.  $Y$  in this case), thus taking them out of the equation.

ex.

|           | rain = T | rain = F |
|-----------|----------|----------|
| cloud = T | 0.5      | 0.05     |
| cloud = F | 0.15     | 0.30     |

| cloud = T | 0.55 |
|-----------|------|
| cloud = F | 0.45 |

$P(\text{cloudy}, \text{rain})$

marginalize out "rain"

$$P(\text{cloudy}) = \sum_{\text{rain} \in \{T, F\}} P(\text{cloudy}, \text{rain})$$

- Independence: when one random variable's probability doesn't depend on another.

$$P(X, Y) = P(X|Y) P(Y) \quad \leftarrow \text{via product rule}$$

$$= P(X) \cdot P(Y) \quad \leftarrow \text{if } X \text{ \& } Y \text{ are independent } P(X|Y) = P(X)$$

ex.  $P(\text{cloudy}, \text{heads}) = P(\text{cloudy}) P(\text{heads})$

- conditional independence: if  $X$  and  $Y$  are conditionally independent given  $Z$  then:

$$P(X, Y|Z) = P(X|Y, Z) \cdot P(Y|Z) \quad \leftarrow \text{Product Rule}$$

$$= P(X|Z) \cdot P(Y|Z) \quad \leftarrow \text{if } X \text{ \& } Y \text{ are indep. given } Z \text{ then: } P(X|Y, Z) = P(X|Z)$$

⇒ once we know  $Z$ , knowing  $Y$  doesn't tell me anything new abt.  $X$ .

ex.  $P(\text{traffic}, \text{umbrella} | \text{rain}) = P(\text{traffic} | \text{rain}) \cdot P(\text{umbrella} | \text{rain})$

**BUT**  $P(\text{traffic}, \text{umbrella}) \neq P(\text{traffic}) P(\text{umbrella})$   
 b/c these two are correlated via the presence/absence of rain!

## BAYES THEOREM

$$P(X|Y) \Rightarrow P(\text{action} | \text{goal})$$

ex. action

ex. goal

What if we know the probability of  $X$  given  $Y$ , but we actually want to know probability of  $Y$  given  $X$ ?

$$P(Y|X) \Rightarrow P(\text{goal} | \text{action})$$

Bayes' Theorem lets us establish a relationship btwn these two!

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

Labels in diagram:  
 -  $P(Y|X)$ : goal (pointing to Y), action (pointing to X)  
 -  $P(X|Y)$ : action (pointing to X), goal (pointing to Y)  
 -  $P(Y)$ : goal (pointing to Y)  
 -  $P(X)$ : action (pointing to X)

② How do we derive it?

$$P(X, Y) = P(Y|X) \cdot P(X) \leftarrow \text{product rule}$$

⇒  
divide by  
 $P(X)$

$$\frac{P(X, Y)}{P(X)} = P(Y|X)$$

⇒  
product  
rule on  
numerator

$$\frac{P(X|Y) \cdot P(Y)}{P(X)} = P(Y|X) \quad \checkmark$$

Each component of Bayes' Rule has a name:

likelihood function  $\equiv$  observation model

posterior

$$P(Y|X) =$$

$$P(X|Y) \cdot P(Y)$$

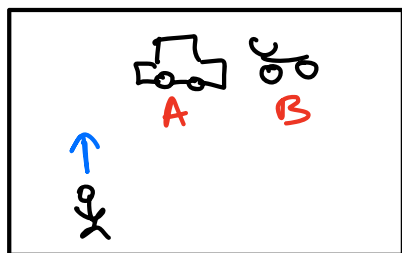
prior

$$P(X)$$

normalize

This rule ends up being very useful for things like state estimation : inferring properties of human behavior!

Example: we want to infer which vehicle a pedestrian wants to move to (A or B). Let's use Bayes' Thm!



PRIOR

$$A = \{\uparrow, \rightarrow\}$$

|          |          |
|----------|----------|
| goal = A | goal = B |
| 0.5      | 0.5      |

$P(\text{goal})$

OBSERVATION MODEL

$P(a | \text{goal})$

$$\begin{aligned} P(a = \uparrow | g = A) &= 0.8 \\ P(a = \rightarrow | g = A) &= 0.2 \\ P(a = \uparrow | g = B) &= 0.4 \\ P(a = \rightarrow | g = B) &= 0.6 \end{aligned}$$

YOU OBSERVE  $a = \uparrow$

② COMPUTE POSTERIOR:  $P(g | a = \uparrow) = ?$

$$P(\text{goal} | a = \uparrow) = \frac{P(a = \uparrow | \text{goal}) P(\text{goal})}{P(a = \uparrow)} \quad // \text{Bayes' Rule}$$

$$= \frac{P(a = \uparrow | \text{goal}) P(\text{goal})}{\sum_{\text{goal} \in \{A, B\}} P(a = \uparrow, \text{goal})} \quad // \text{marginalize over goals}$$

$$= \frac{P(a = \uparrow | \text{goal}) P(\text{goal})}{\sum_{\text{goal} \in \{A, B\}} P(a = \uparrow | \text{goal}) P(\text{goal})} \quad // \text{product rule}$$

$P(\text{goal})$

$\swarrow$  prior

| goal = A | goal = B |
|----------|----------|
| 0.5      | 0.5      |

$P(a | \text{goal})$

|                                              |
|----------------------------------------------|
| $P(a = \uparrow   \text{goal} = A) = 0.8$    |
| $P(a = \rightarrow   \text{goal} = A) = 0.2$ |
| $P(a = \uparrow   \text{goal} = B) = 0.4$    |
| $P(a = \rightarrow   \text{goal} = B) = 0.6$ |

$P(\text{goal} | a = \uparrow)$

$\nwarrow$  posterior

| goal = A | goal = B |
|----------|----------|
| 0.66     | 0.33     |

goal A:

$$P(\text{goal} = A | a = \uparrow) = \frac{P(a = \uparrow | \text{goal} = A) P(\text{goal} = A)}{P(a = \uparrow | \text{goal} = A) P(\text{goal} = A) + P(a = \uparrow | \text{goal} = B) P(\text{goal} = B)}$$

$\swarrow 0.8 \quad \swarrow 0.5 \quad \swarrow 0.4 \quad \swarrow 0.5$

$$= 0.66$$

goal B:

$$P(\text{goal} = B | a = \uparrow) = 1 - P(\text{goal} = A | a = \uparrow) = 0.33 \quad \leftarrow \text{law of total prob.}$$

KEY IDEA: Humans act in ways driven by their goals!

↳ robot can use this model to infer the H's goal/intent...

Q Where exactly does this probability model  $P(X|Y)$  come from?

A Cognitive psychology, econometrics, data-driven modeling, common sense/heuristic  
↳ application of statistical methods for economic relations

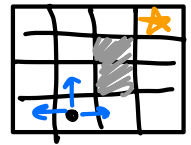
Boltzmann Rational Model [Luce 1959, 1977; Ziebart 2010]

Assume that people are optimizing some reward function in their mind (e.g. dist to goal). This model says that people behave approximately optimally (i.e. approx. rationality) in pursuit of their goals:

action ( $\leftarrow, \uparrow, \rightarrow, \downarrow$ )

$$P(\uparrow | s; \text{goal}) = \frac{e^{Q(s, \uparrow; g)}}{\sum_{\bar{a} \in A} e^{Q(s, \bar{a}; g)}}$$

$\bar{a} \in A \leftarrow$  set of all actions



⇒ the higher the value of an action  $a$ , the exponentially more likely it is that the human will take this action

Another variation of this model you will see a lot in the class is a "trajectory-level model":

let trajectory be:  $\zeta = (s_0, a_0, s_1, a_1, \dots)$

cumulative reward  $R(\zeta) := \sum_{(s,a) \in \zeta} r(s,a)$

$$P(\zeta | \text{goal}) = \frac{e^{R(\zeta; \text{goal})}}{\sum_{\bar{\zeta} \in \Xi} e^{R(\bar{\zeta}; \text{goal})}}$$

$\bar{\zeta} \in \Xi \leftarrow$  set of all trajectories

